

MIAE 215 · Mini-Course Overview & Guidelines

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering (MIAE)

Source: [Getting Started C++ Mini-Course](#)

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1. Course Purpose & The 2-Week Head Start

This online mini-course is designed to get engineering students programming with C++ in a short period of time (2 weeks). This "head start" provides students with a solid background to learn more advanced C++ topics covered in **MIAE 215** (also known as **MECH 215**) and hit the ground running in a busy first term.

A large number of students fall behind in MIAE 215 due to an insufficient computer and programming background. The small amount of time you invest (about **5 to 10 hours total**) with this mini-course could save you dozens of hours of struggle later in the semester.

Important Academic Notice:

There are **no labs or tutorials during the first three weeks of the term**. Instead, students are expected to complete Lessons #1 through #4 during the first week, and all remaining lessons (#5 and #6) by the end of the second week. Regular MIAE 215 lectures strictly assume you have completed this mini-course!

2. Why Mechanical, Aerospace & Industrial Engineers Need C++

C++ programming is a vital core skill for modern engineers:

- **The Language of Automation:** Computers execute industrial automation, CNC machining, robotic assembly lines, and flight control systems.
- **Mechatronics & Embedded Systems:** Departmental project courses such as **MECH 390** (mini-capstone involving mobile robots) and **MECH 490** (capstone mechatronics projects) heavily rely on Arduino and ARM microcontroller boards programmed in C++.
- **Competitive Clubs & Society Teams:** Engineering teams including **Space Concordia** (rocketry, satellites, rovers) and **SAE Concordia** (Formula racing, aero design) write high-performance telemetry, flight software, and sensor acquisition loops in C++.

3. Schedule, Milestones & Time Commitment

RECOMMENDED 2-WEEK TIMELINE		
Period	Lessons to Complete	Target Hours
Week 1	Lesson 1: Software Install	1.5 – 2.0 hours
	Lesson 2: C++ Programs	1.5 – 2.0 hours
	Lesson 3: Variable Types	1.5 – 2.0 hours
	Lesson 4: Expressions	1.5 – 2.0 hours
Week 2	Lesson 5: Control Flows	1.5 – 2.0 hours
	Lesson 6: Arrays	1.5 – 2.0 hours
Total Estimated Investment: 8 – 10 Hours		

Key Advice from Prof. Gordon:

Roughly 1–2 hours should be spent on each lesson. It is **not recommended** to attempt the entire course in 1 day or 1 week; it takes time for mental models of memory and syntax to settle. If you finish early, work through the additional exercises provided in each lesson folder!

4. Official Communication & Support Protocol

- **Lessons #1 & #2 Support:** If you encounter installation or compiler setup issues that cannot be resolved via the Troubleshooting Guide, you may contact Prof. Gordon at brandon.gordon@concordia.ca with the subject line "**MIAE 215 mini course**".
- **Lessons #3 through #6:** These lessons are structured for self-directed mastery. Use the comprehensive step-by-step solutions provided in each lesson folder to verify your code.